Hidden in the Leftovers

Task-specific reconstruction gaps of sparse autoencoders, in English and Indian languages

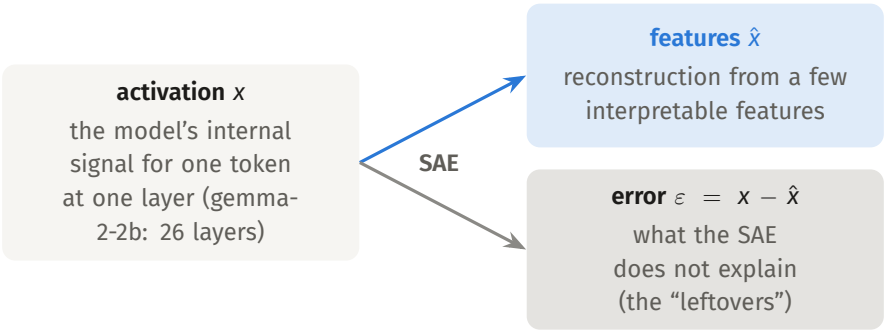
Prakhar Singhal (2022111025) Raveesh Vyas (2022114007)

Team IpCp--

October 2026

Mechanistic Interpretability of Language Models (CL3.412)

Background · SAEs split activations into features and error



Sparse autoencoder (SAE): the standard tool for decomposing activations into interpretable features [1, 8].

Reconstruction fidelity: $\hat{x}$ in place of x keeps **98–99%** of next-token performance.

Background · Does task computation reside in the SAE error?

- The error is large and partly structured [4], and matters in circuits [12].
- If task-relevant state sits in ε, feature-based explanations are incomplete.
- Fidelity is an average over generic text; it does not test a specific computation.

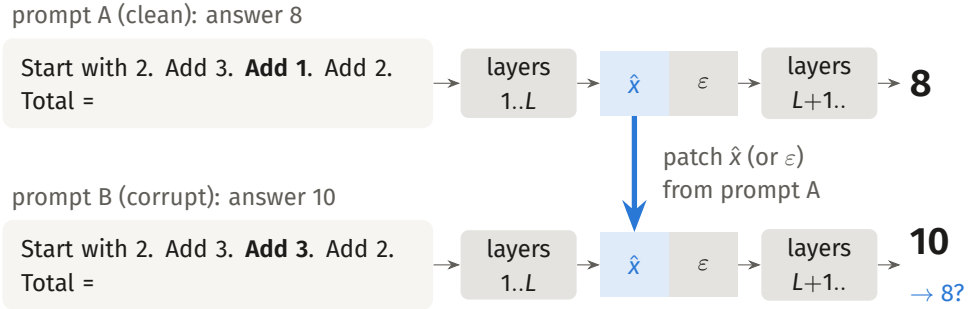
Pilot question: for state-tracking tasks, is the causal effect carried by $\hat{x}$ or by ε?

features $\hat{x}$
measured by fidelity

error ε
not measured per task

?

Method · Activation patching on prompt pairs



Effect recovered (NLDR): normalized logit difference, 0 = B's answer, 1 = A's answer. Patching all of x gives 1 by construction [16].

Pilot · Four tasks the model performs reliably

Running sum

`Start with 2. Add 3. Add 1. Add 2. Total =` **8**

tracks a total

Object tracking

`The key is in box B. … The key moves to box C. … The key is in box` **C**

entity–box binding [9]

Variable copying

`t = 7. q = 4. x = q. q = 2. x =` **4**

follows an assignment

Names (IOI)

control

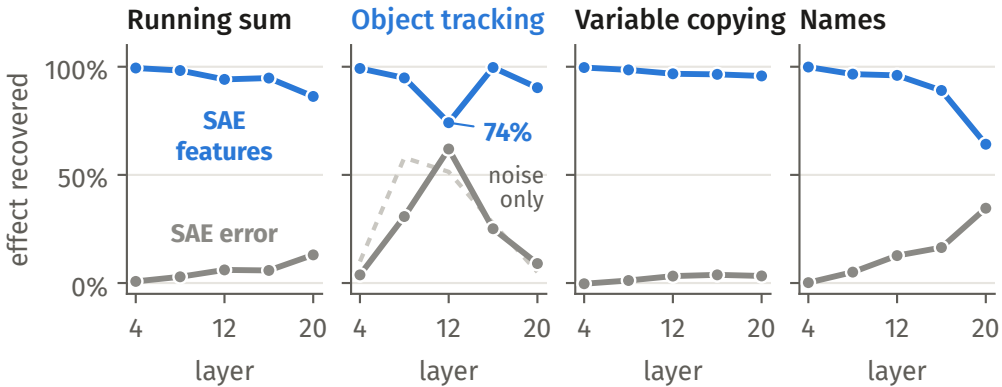
`When Mary and John went to the store, John gave a drink to` **Mary**

known circuit [15]

gemma-2-2b [6] · Gemma Scope residual SAEs [10] · 150 test pairs per task

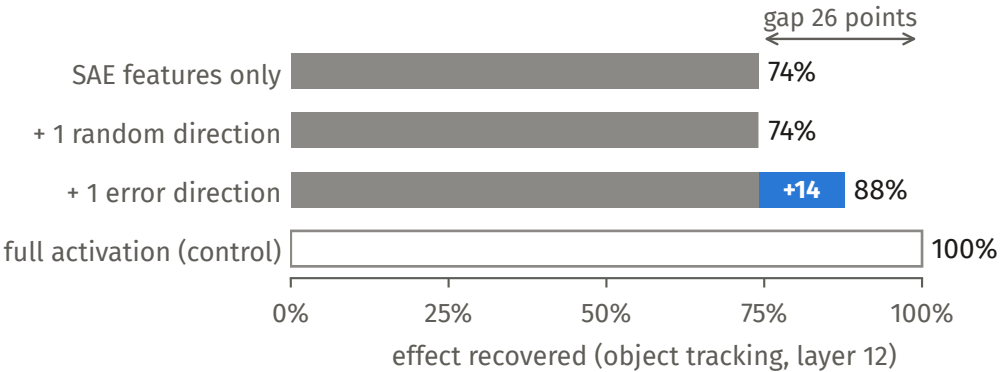
Three planned algorithmic tasks were dropped (shortcut-level accuracy).

Result 1 · SAE features carry 74–100% on tracking tasks



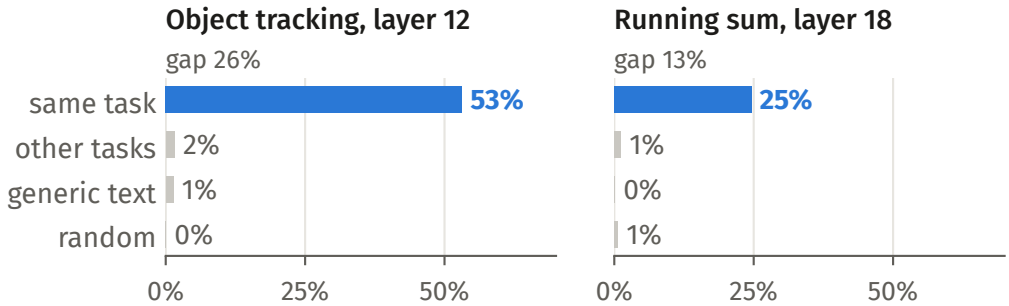
Error alone: 0–13%; object tracking's bump matches a noise control (dashed).

Result 2 · One error direction closes half the gap (L12)



+14 points = 52% of the gap. **Error direction:** top principal direction of ε.

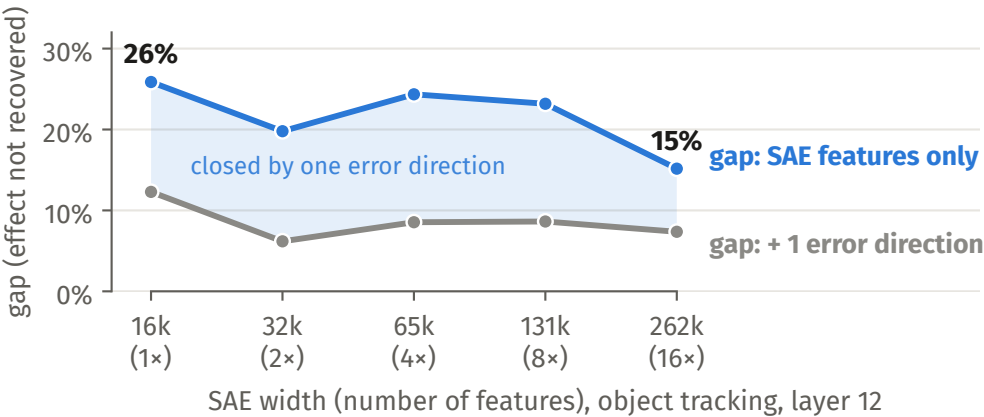
Result 3 · The recovering directions are task-specific



share of the gap closed by 2 directions fitted on each source

Directions fitted on other tasks, generic text (the Pile) or at random close at most 2%.

Result 4 · A 16× wider SAE reduces the gap from 26% to 15%



One error direction closes 51–69% of the remainder; width and sparsity confounded.

Proposal · Hypothesis and research questions

Hypothesis: some SAE reconstruction gaps are concentrated in a few task-specific error directions, and average fidelity does not detect them.

Q1

Which tasks?

size and specificity of the gap across task families

Q2

What mechanism?

is the error direction part of entity binding?

Q3

Which languages?

is the gap larger in Hindi and Hinglish?

Plan · Q1 and Q2: task families and the binding test

Task family	Tasks	Status
Entity binding	objects in boxes; people and cities	pilot + new
Counting	running sum; longer chains	pilot + new
Reference following	variable copying	pilot
Control	names (IOI)	pilot

Q1: gap, directions needed to close half, specificity; widths compared at matched sparsity.

Criterion: gap $\geq$ 10%; same-task directions close $\geq$ 30 points more than any other source.

Q2: binding-ID vectors were found in LLaMA and Pythia [5]; re-derive them for gemma-2-2b, then edit along the error direction.

Criterion: $\geq$ 50% of pairs switch to the other box, against $\leq$ 5% for a random direction.

Plan · Q3: the same analysis in Indian languages

English (pilot baseline)

Hindi

Hinglish (Latin script)

Tamil or Bengali (if feasible)

French (non-Indian control)

Gemma Scope SAEs were trained on predominantly English text [10]; prior work studies language-specific features [2], not per-language error.

Measures per language

1. Reconstruction fidelity on Hindi vs English text
2. Patching gap and its specificity
3. Transfer of the English error direction to Hindi

Criterion: Hindi gap larger, non-overlapping 95% CIs.

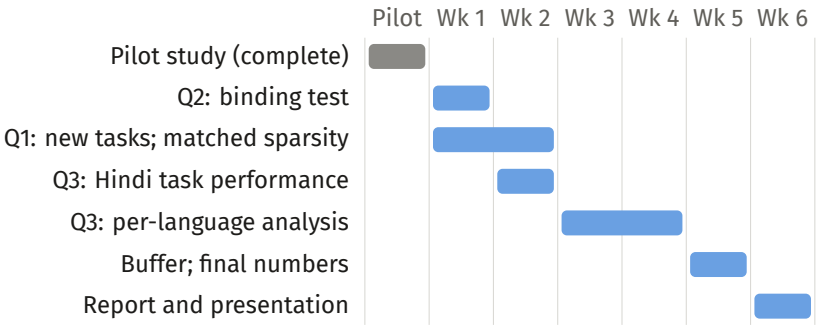
Translated prompts (draft, to be validated)

शुरुआत 2 से करें। 3 जोड़ें। 1 जोड़ें। 2 जोड़ें। कुल = **8**

Rahul aur Priya market gaye; Priya ne book di. Book kisko mili? **Rahul**

Data: IndicCorp v2 [3], FineWeb-2 [13].

Plan · Timeline and deliverables



Deliverables: a table of gaps by task, layer, width and language · the patching code · a result on the binding hypothesis

Compute: the pilot ran on one 8 GB GPU in about 6 h.

Summary

	Pilot result	**Next step**
Where is the effect?	features carry 74–100% on tracking tasks	–
Is the gap concentrated?	object tracking (L12): 1 direction closes 52% of a 26% gap	Q1: other task families
Is it task-specific?	other tasks, generic text: $\leq 2\%$	Q1: transfer within a family
Does width remove it?	26% $\rightarrow$ 15% at 16$\times$ width	Q1: matched sparsity
Mechanism?	–	Q2: binding test
Other languages?	–	Q3: Hindi, Hinglish

References

[1] T. Bricken et al. Towards monosemanticity: Decomposing language models with dictionary learning. Transformer Circuits Thread, 2023. Published October 4, 2023.

[2] B. Deng et al. Unveiling language-specific features in large language models via sparse autoencoders. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 4563–4608, July 2025. Association for Computational Linguistics.

[3] S. Doddapaneni et al. Towards leaving no Indic language behind: Building monolingual corpora, benchmark and models for Indic languages. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 12402–12426, July 2023. Association for Computational Linguistics.

[4] J. Engels, L. R. Smith, and M. Tegmark. Decomposing the dark matter of sparse autoencoders. *Transactions on Machine Learning Research*, 2025.

[5] J. Feng and J. Steinhardt. How do language models bind entities in context? In *The Twelfth International Conference on Learning Representations (ICLR)*, 2024.

[6] Gemma Team et al. Gemma 2: Improving open language models at a practical size. *arXiv preprint arXiv:2408.00118*, 2024.

[7] J. Hewitt and P. Liang. Designing and interpreting probes with control tasks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 2733–2743, Nov. 2019. Association for Computational Linguistics.

[8] R. Huben et al. Sparse autoencoders find highly interpretable features in language models. In *The Twelfth International Conference on Learning Representations (ICLR)*, 2024.

References

[9] N. Kim and S. Schuster. Entity tracking in language models. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3835–3855, July 2023. Association for Computational Linguistics.

[10] T. Lieberum et al. Gemma Scope: Open sparse autoencoders everywhere all at once on Gemma 2. In *Proceedings of the 7th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP*, pages 278–300, Nov. 2024. Association for Computational Linguistics.

[11] A. Makelov et al. Is this the subspace you are looking for? an interpretability illusion for subspace activation patching. In *The Twelfth International Conference on Learning Representations (ICLR)*, 2024.

[12] S. Marks et al. Sparse feature circuits: Discovering and editing interpretable causal graphs in language models. In *The Thirteenth International Conference on Learning Representations (ICLR)*, 2025.

[13] G. Penedo et al. FineWeb2: One pipeline to scale them all — adapting pre-training data processing to every language. In *Second Conference on Language Modeling (COLM)*, 2025.

[14] S. Rajamanoharan et al. Jumping ahead: Improving reconstruction fidelity with JumpReLU sparse autoencoders. *arXiv preprint arXiv:2407.14435*, 2024.

[15] K. R. Wang et al. Interpretability in the wild: a circuit for indirect object identification in GPT-2 small. In *The Eleventh International Conference on Learning Representations (ICLR)*, 2023.

[16] F. Zhang and N. Nanda. Towards best practices of activation patching in language models: Metrics and methods. In *The Twelfth International Conference on Learning Representations (ICLR)*, 2024.

Questions

Prakhar Singhal · prakhar.singhal@research.iiit.ac.in
Raveesh Vyas · raveesh.vyas@research.iiit.ac.in

Team IpCp-- · CL3.412

Backup slides

B1 · Setup and measures

Setup

- gemma-2-2b, 26 layers, $d = 2304$
- Gemma Scope JumpReLU SAEs [14], widths 16k–262k
- 300 train / 150 test pairs per task, prompt-disjoint
- 95% paired bootstrap CIs
- 78 unit tests; top-1 agreement with Hugging Face 95%

Measures

- NLDR $= \frac{D_{\text{patched}} - D_B}{D_A - D_B}$, D: logit difference
- Gap $= 1 -$ NLDR($\hat{x}$ only)
- Repair: $\hat{x} + U_r U_r^\top \varepsilon$, U_r fitted on training pairs
- Controls: random directions; ε replaced by noise
- Fidelity: CE loss recovered, 16k SAEs, layers 10–18, the Pile

B2 · Three methodological pitfalls

Patch position

47% vs 100%

object tracking, layer 12: last token only vs all tokens from the first difference

Probe baselines

100%

a probe on prompt tokens alone solves variable copying and names [7]

Decodable $\neq$ used

84% vs 47%

probe accuracy on ε (chance 18%) vs effect of patching that token

B3 · Original hypotheses and pilot verdicts

Original hypothesis	Verdict
H1: ε carries the task state	**Rejected:** $\hat{x}$ carries 74–100% on the three tracking tasks; ε 0–13% (object tracking up to 62%, largely a noise effect)
H2: the state is a low-dimensional structure in ε	**Not supported as stated:** decodable from ε (84%) but not causally used at the last token
H3: a few error directions repair $\hat{x}$	**Supported, narrowly:** one direction closes 52% of object tracking's gap at layer 12; task-specific

B4 · Caveats

- One model, one SAE family, synthetic tasks; width tested at layer 12 only.
- Two kinds of gap: IOI loses 24–36% at layers 18–20, spread over $\sim$32 directions (layer 18).
- Object tracking's error-only 62% is inflated: noise in place of ε gives 51%.
- Active features per token: 124–139 at 131k/262k vs 78–89 at 16k; width and sparsity confounded.
- Pre-set persistence rule (widest SAE keeps $\geq$ 60% of the gap) narrowly missed: 15.2% vs 15.5%.
- Low-rank interventions can create interpretability illusions [11]; random and generic-text controls and held-out pairs address this only in part.